Auteur: Mathijs Pittoors
Studierichting: Informatica
Oefeningenreeks 5L nr. 1.3

$$S = 2\pi \int_0^2 \sqrt{x} \cdot \sqrt{1 + \left(\frac{1}{2\sqrt{x}}\right)^2} dx$$

Tekening

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$$= 2\pi \int_0^2 \sqrt{x + \frac{1}{4}} dx$$

Onbepaalde integraal:

Neem $t = x + \frac{1}{4} \Rightarrow dt = dx$

$$\int_0^2 \sqrt{x + \frac{1}{4}} dx = \frac{2}{3} t\sqrt{t} + c = \frac{2}{3}\left(x + \frac{1}{4}\right)\sqrt{x + \frac{1}{4}} + c$$

$$= 2\pi \left[\left(+\frac{1}{4}\right)\sqrt{x + \frac{1}{4}}\right]_0^2$$

$$\frac{4\pi}{3}\left(\left(\frac{9}{4}\sqrt{\frac{9}{4}}\right) - \left(\frac{1}{4}\sqrt{\frac{1}{4}}\right)\right)$$

$$= \frac{13\pi}{3}$$

References

[1] https://www.geogebra.org/m/r4gweqnt